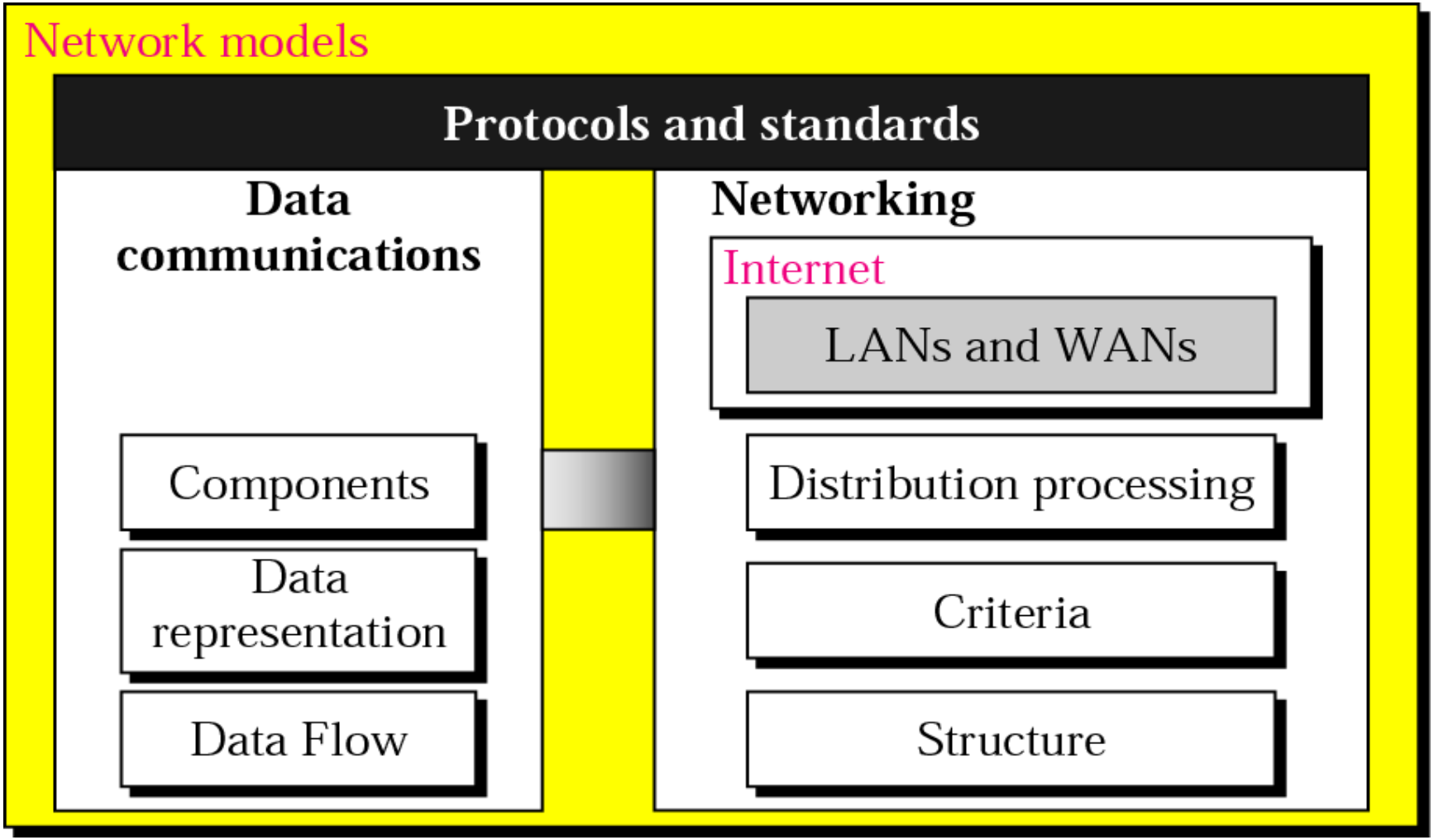
Computer Networks

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Overview


Network models
Protocols and standards
Data communications
Components
Data representation
Data Flow
Networking
Internet
LANs and WANs
Distribution processing
Criteria
Structure

Data (computer) Communications ?

- Exchange of data between two devices via some form of transmission media.

- For data communication communicating devices must be part of a communication system made up of a combination of H/W and S/W.

Effectiveness of DATA communication Depends on

- Delivery
- Accuracy
- Timeliness
- Jitter: Variation in the packet arrival time .

  It is the uneven delay in the delivery of audio and video packets.

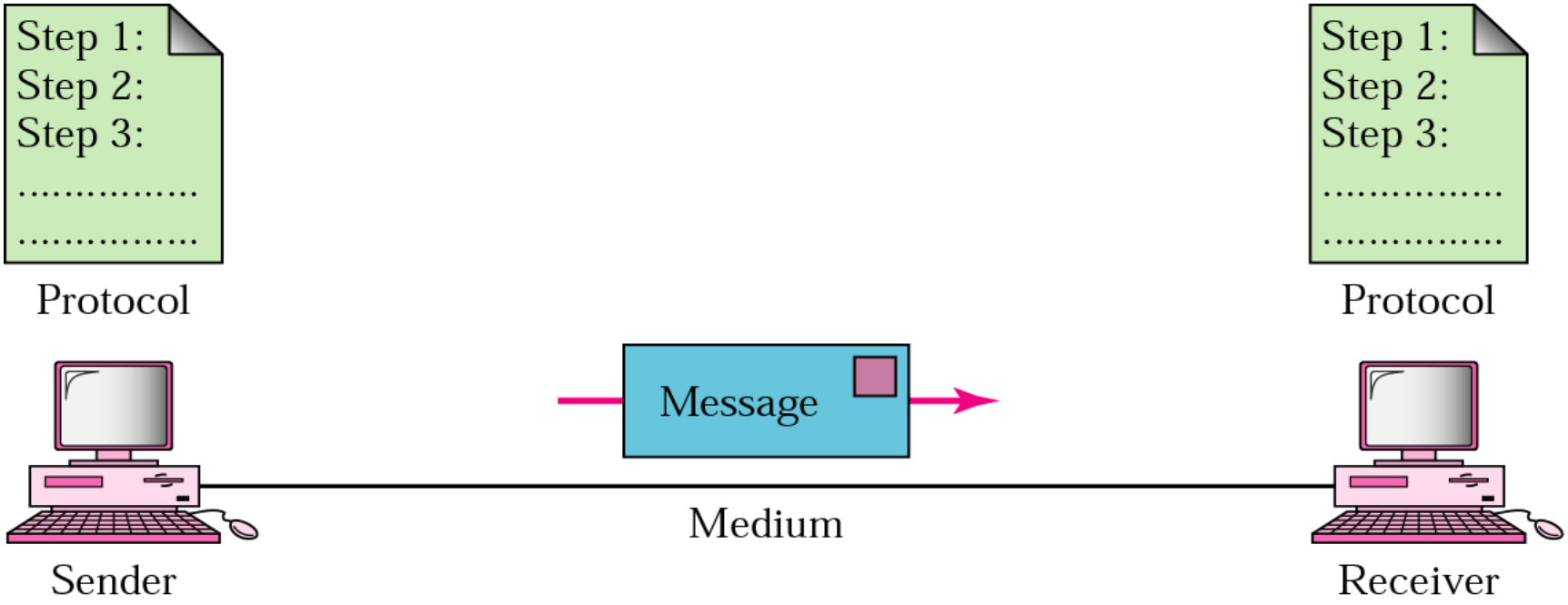
1.1 Data Communication

Components

Data Representation

Direction of Data Flow

Five components of data communication



- **Protocol** :**is a set of rules that govern data communications . It represent an agreement between the communicating devices. Without a protocol ,two devices may be connected but not communicating.**

Figure 1.2 ***Simplex***

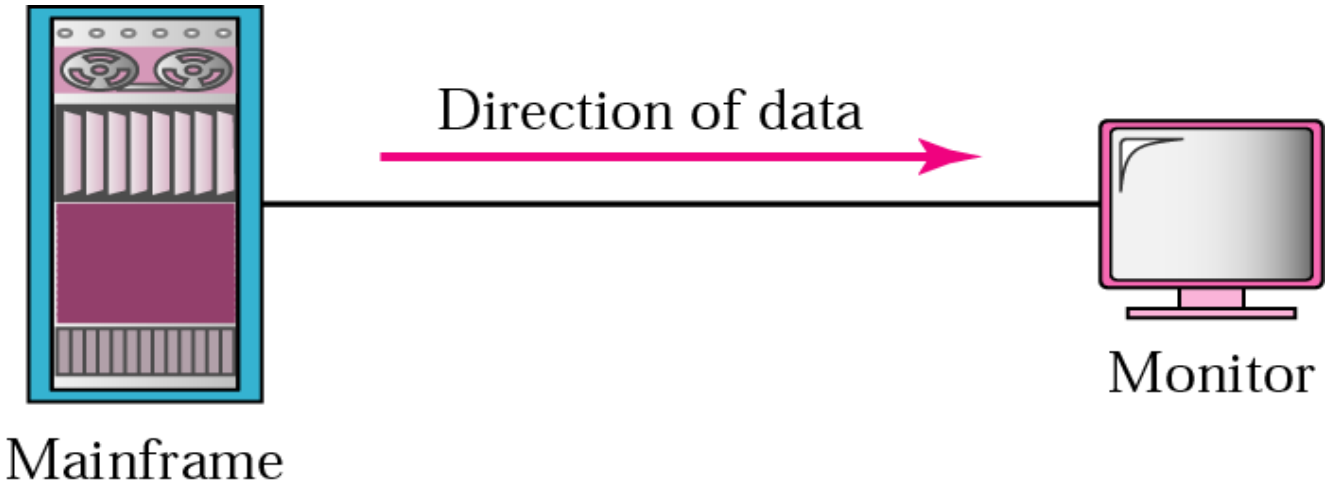


Figure 1.3 ***Half-duplex***

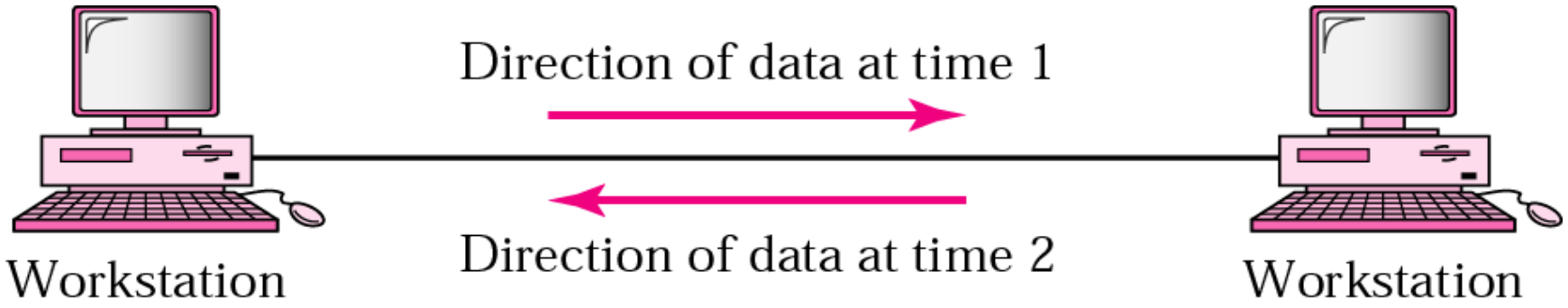
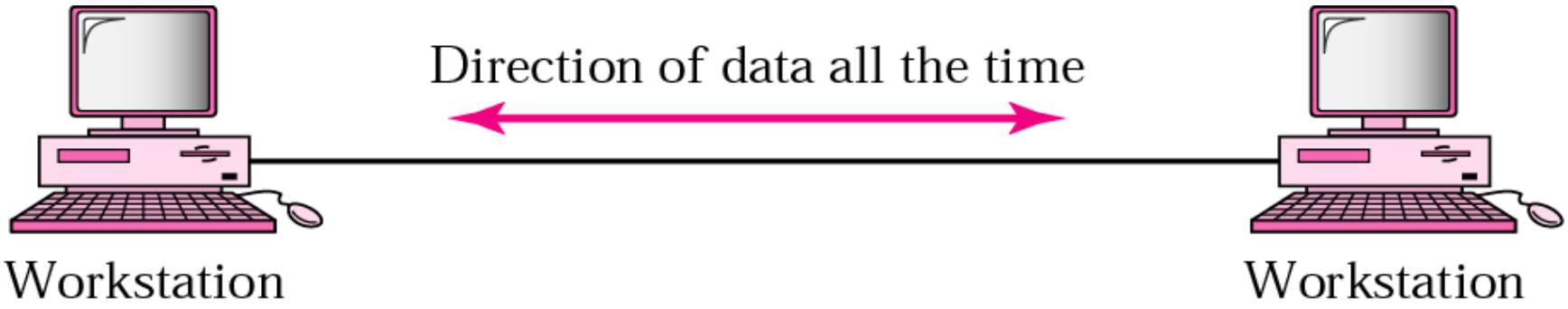


Figure 1.4 *Full-duplex*



1.2 Networks

Distributed Processing

Network Criteria

Physical Structures

Categories of Networks

Figure 1.5 ***Point-to-point connection***

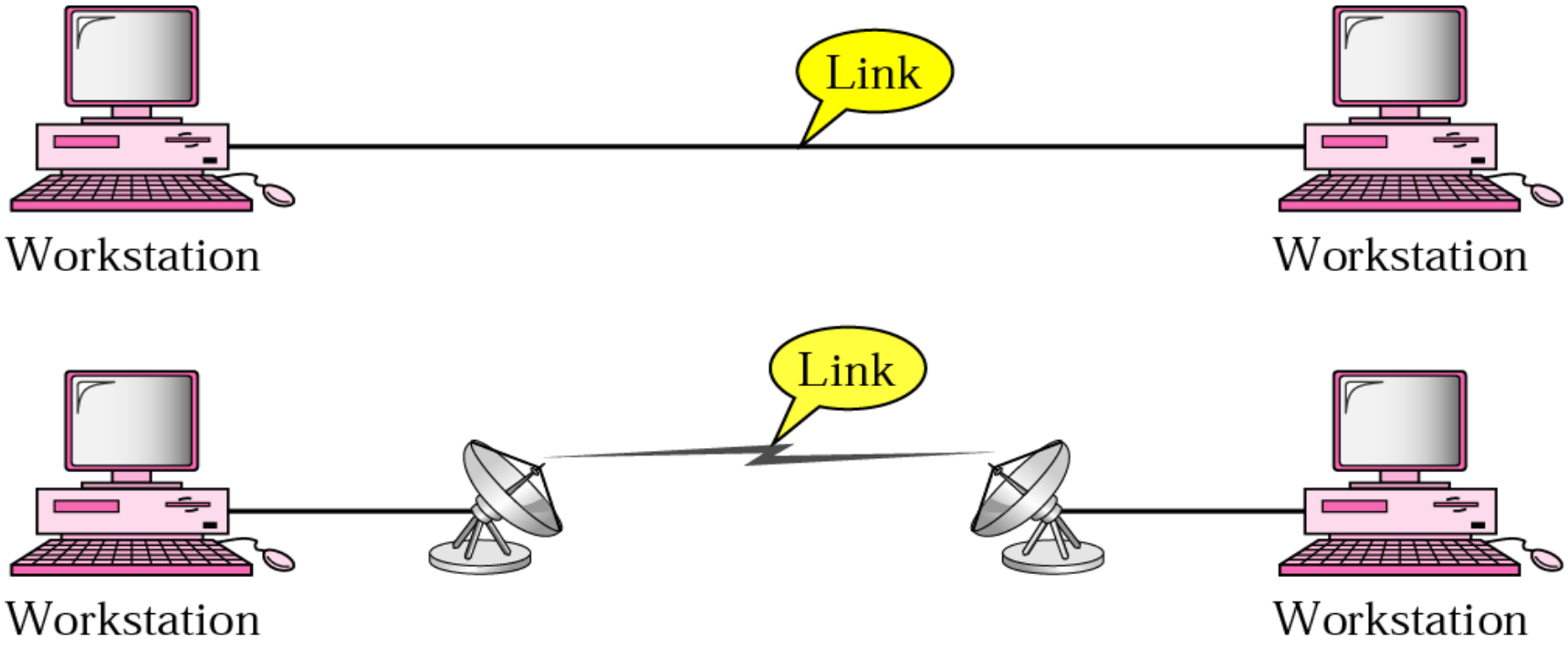


Figure 1.6 *Multipoint connection*

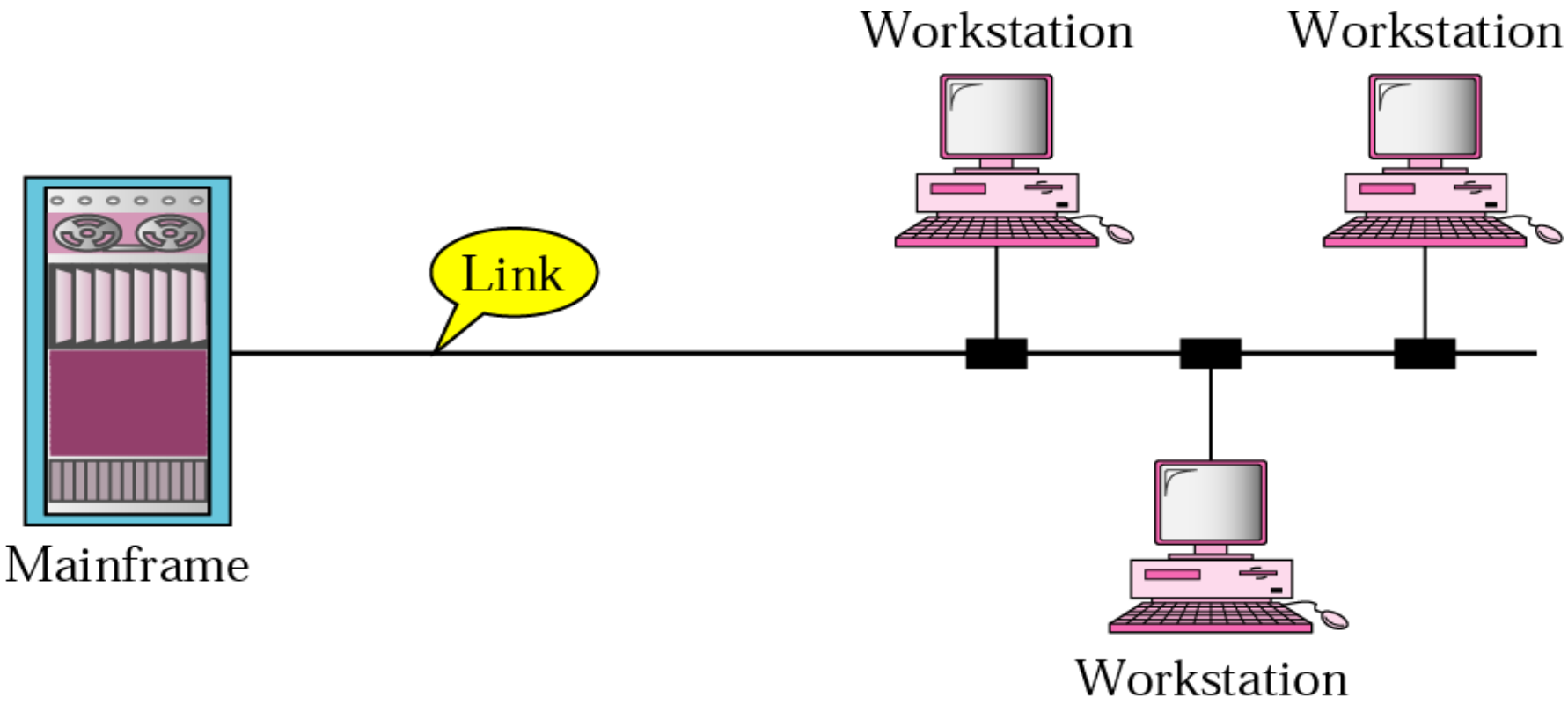


Figure 1.7 *Categories of topology*

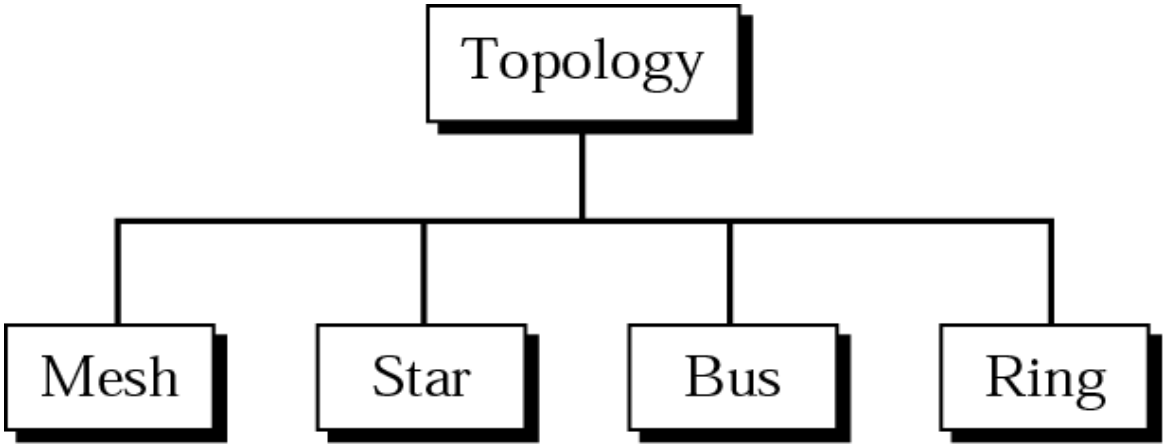


Figure 1.8 *Fully connected mesh topology (for five devices)*

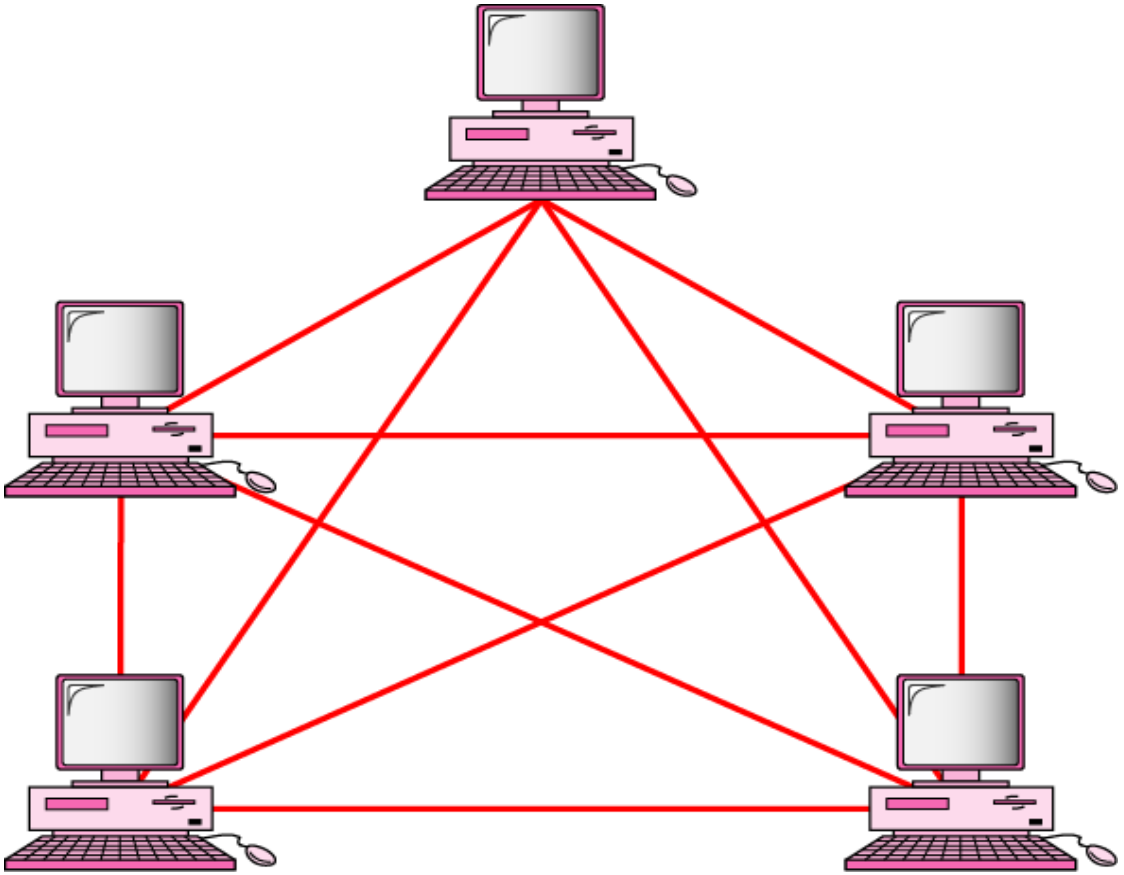


Figure 1.9 *Star topology*

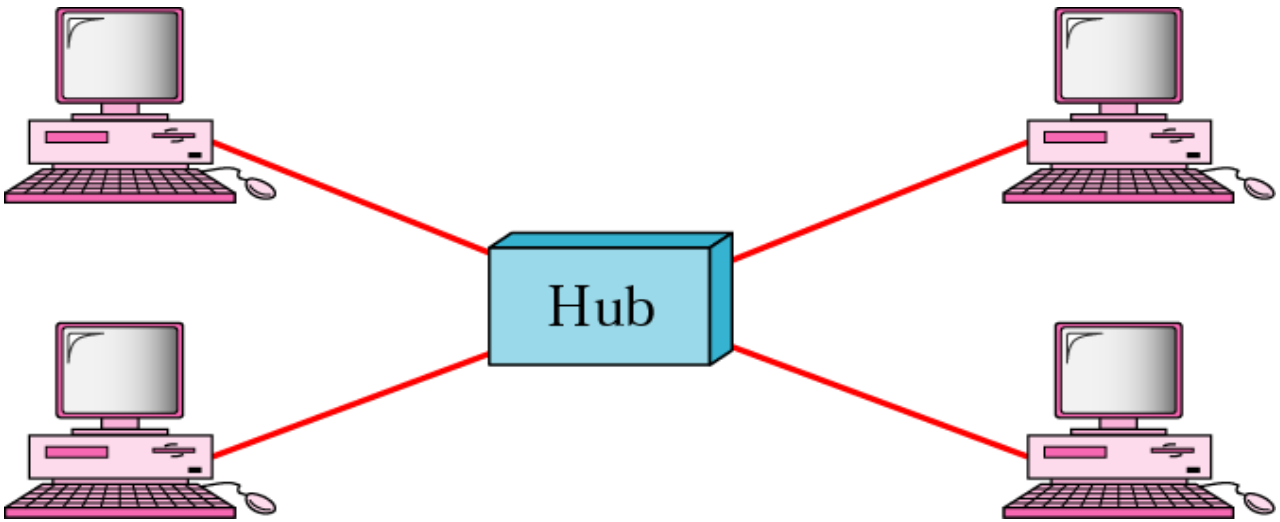


Figure 1.10 *Bus topology*

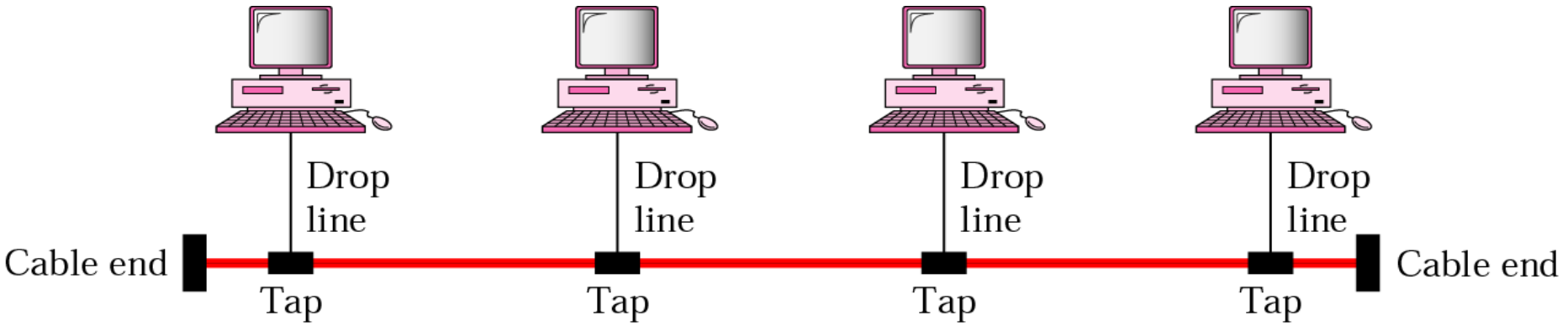


Figure 1.11 *Ring topology*

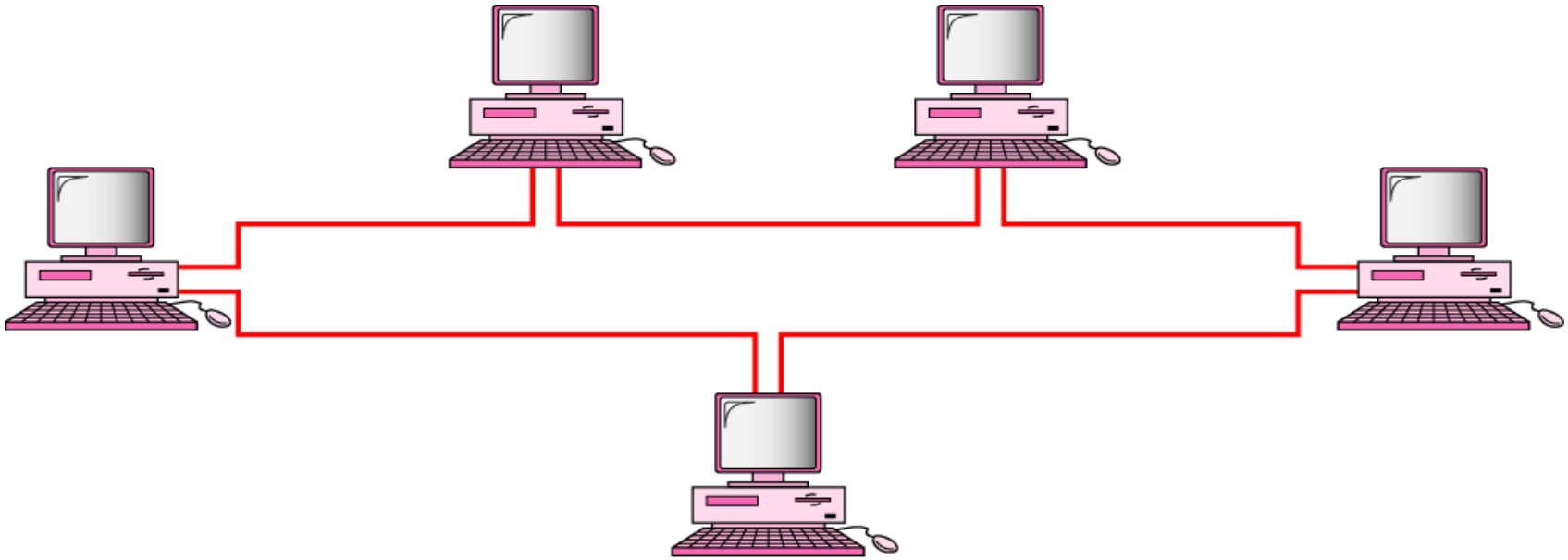


Figure 1.12 *Categories of networks*

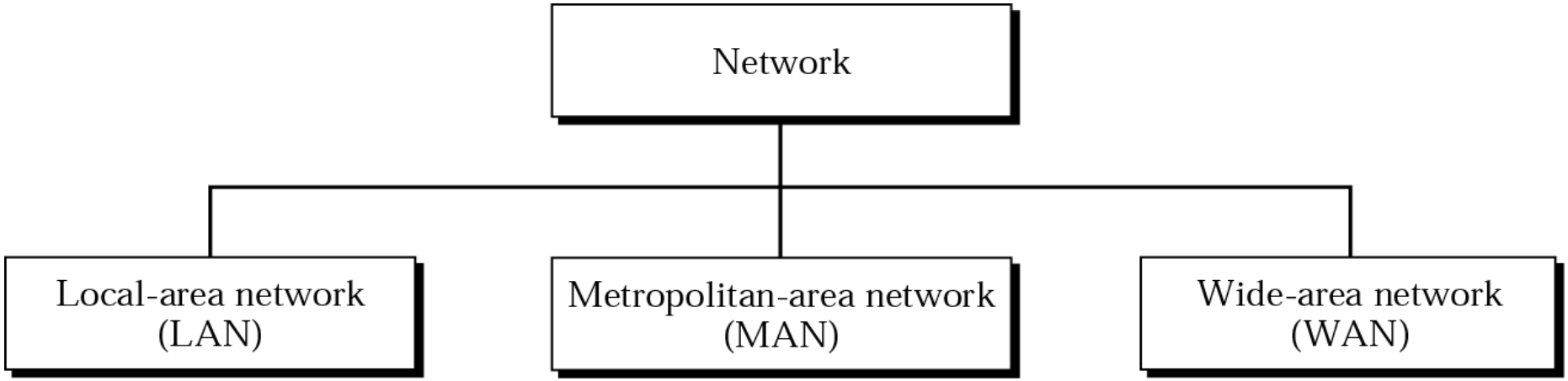
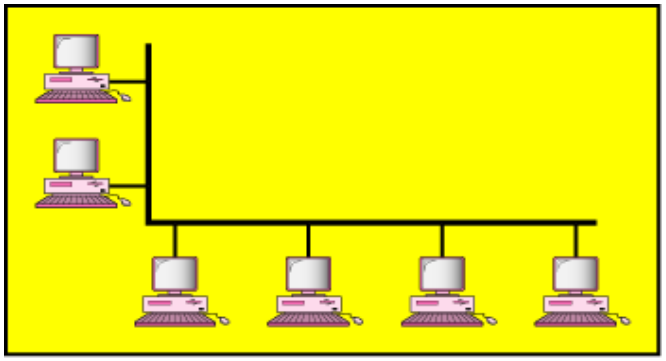
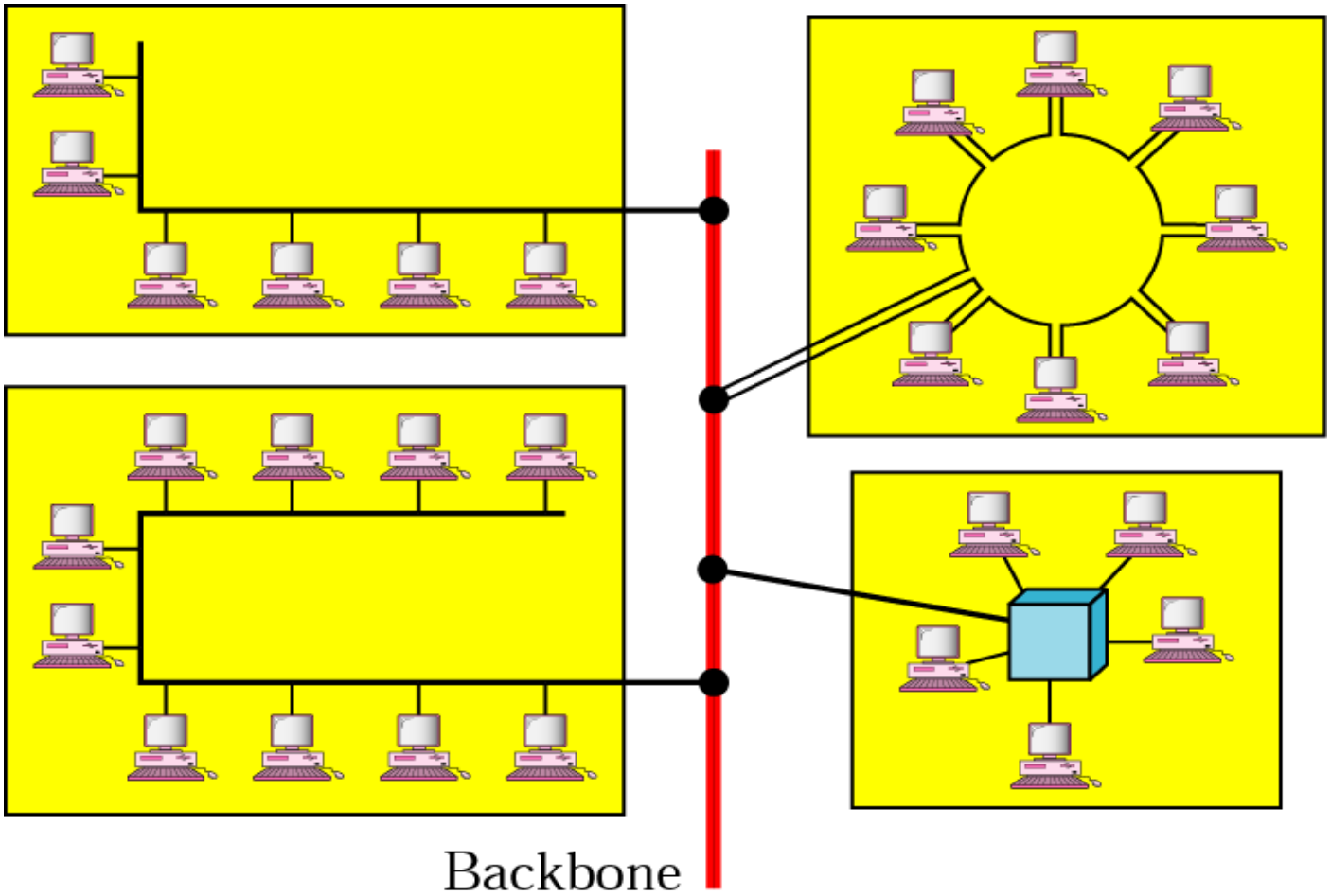


Figure 1.13 *LAN*



a. Single-building LAN

Figure 1.13 *LAN (Continued)*



b. Multiple-building LAN

Figure 1.14 *MAN*

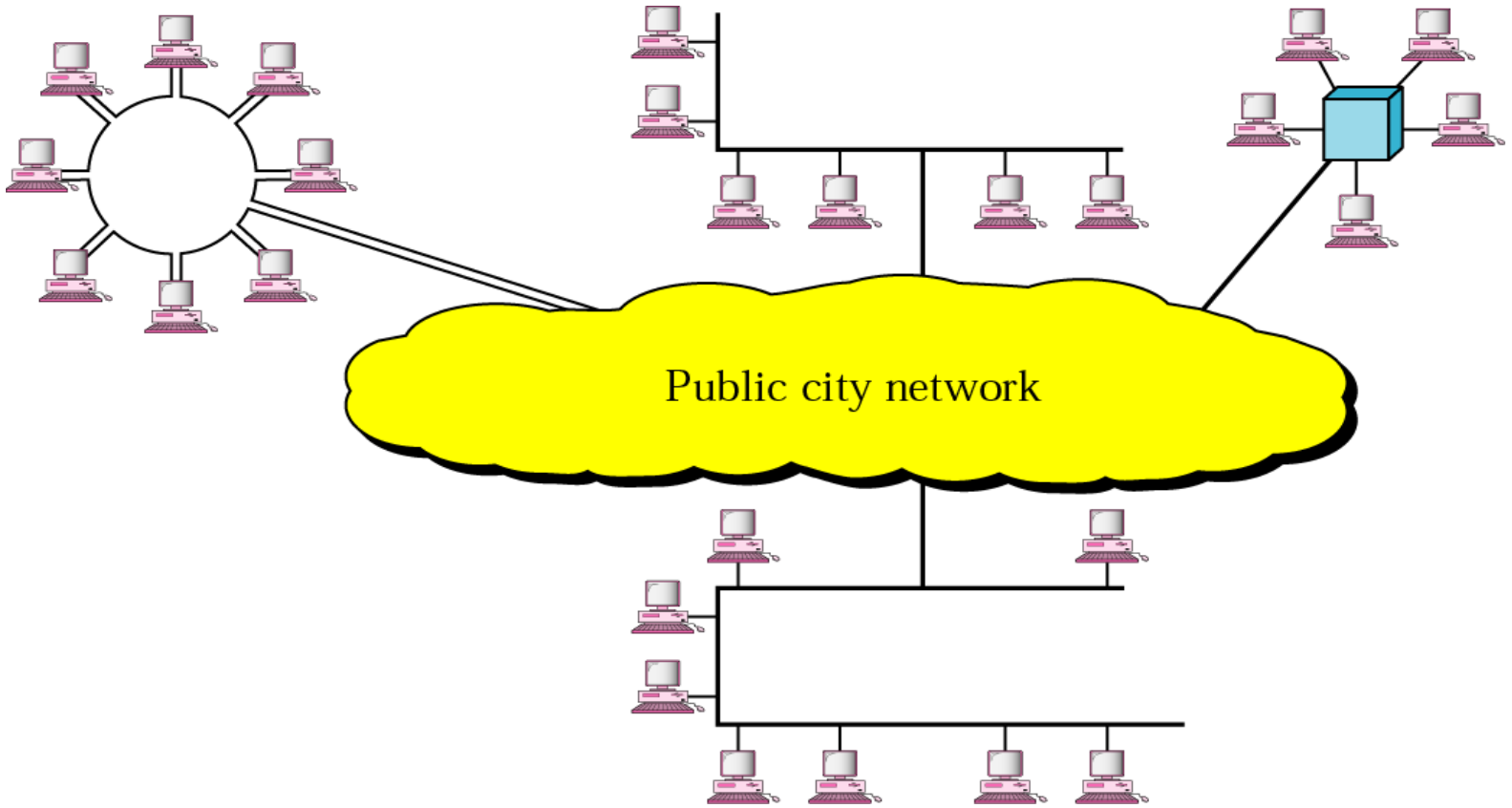
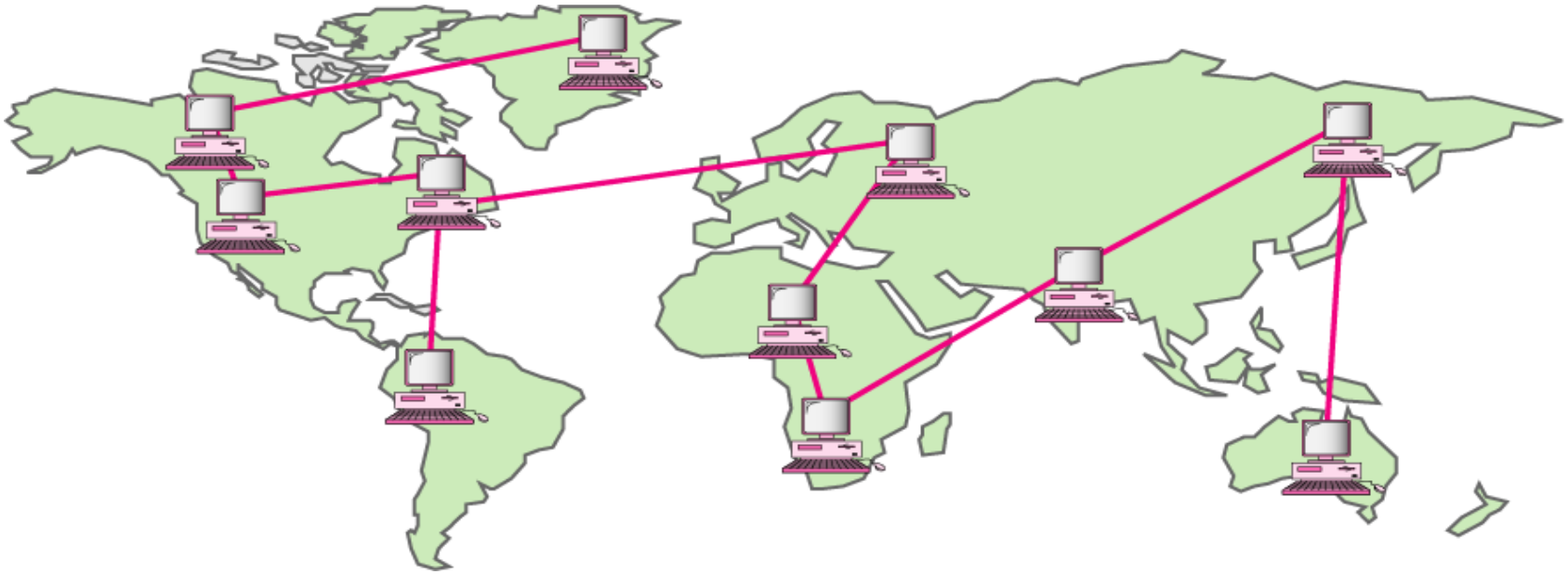


Figure 1.15 *WAN*



1.3 The Internet

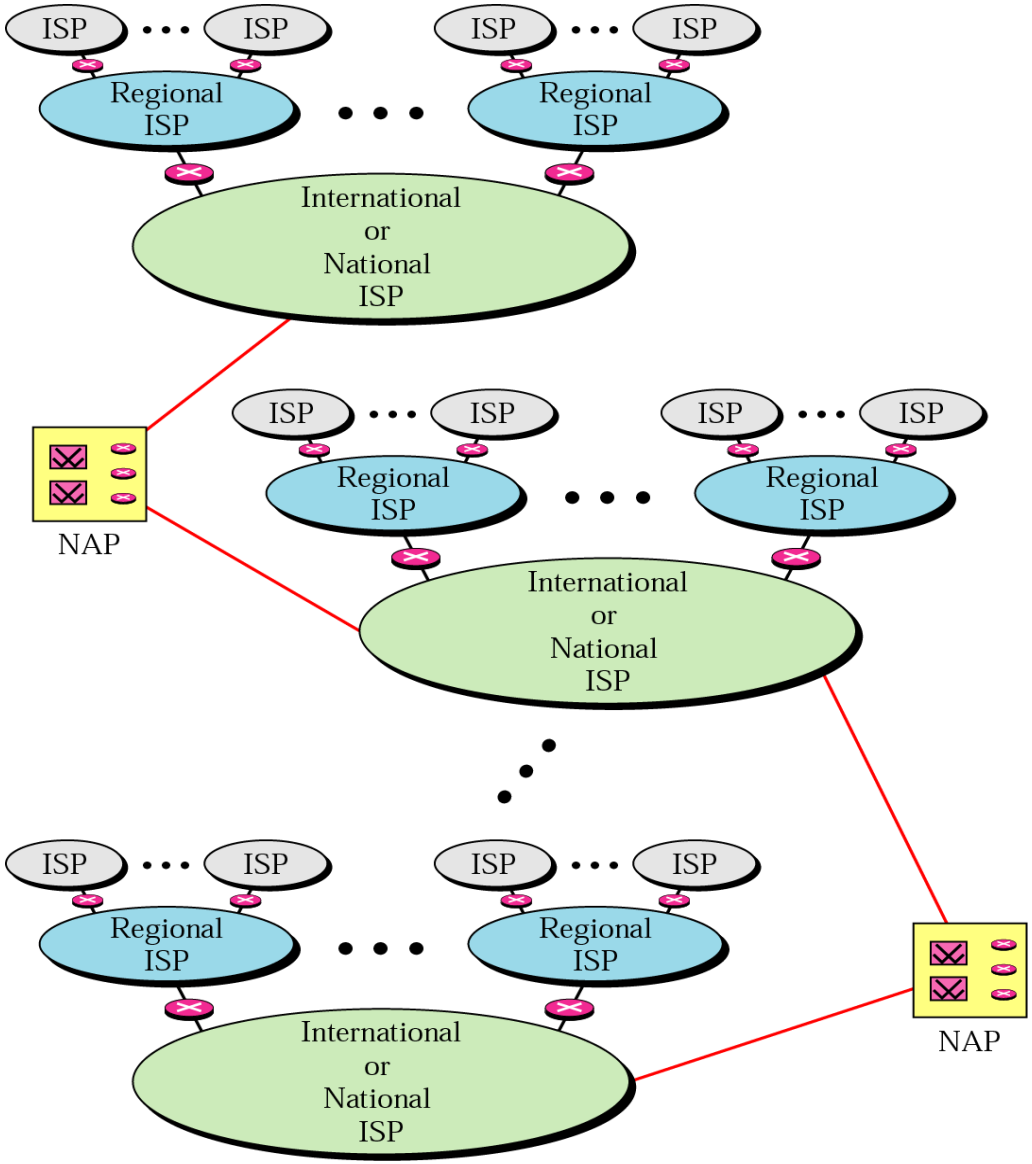
A Brief History

The Internet Today

Internet?

- Network: group of connected communicating devices.
- internet: Two or more networks that can communicate with each other.
- Internet: a collaboration of more than hundreds of thousands of interconnected networks

Figure 1.16 *Internet today*



1.4 Protocols and Standards

Protocols

Standards

Standards Organizations

Internet Standards

Protocol?

- A set of rules that governs data communications. A protocol defines what is communicated, how it is communicated & when it is communicated.

The key elements of Protocols are

- Syntax: Structure or format of the data. meaning the order in which they are present. For e.g. simple protocol might be the first 8 bits of data to be address of the sender the second 8 bits to be the address of the receiver and the rest of the stream to be the message itself
- Semantics: Meaning of each section of bits. How is a particular pattern to be interpreted, and what action is to be taken based on that interpretation? For e.g. Does an address identify the route to be taken or the final destination of the message.
- Timing: When data should be sent and how fast they can be sent. For e.g. If a sender produce data at 100 Mbps but the receiver can process data at only 1 Mbps, the transmission will overload the receiver and some data will lost.

Standards

- It is essential for creating and maintaining an open and competitive market for equipment manufacturers and in guaranteeing national and international interoperability of data and telecommunication technology and processes.

- It also provides guidelines to manufacturers , vendors , government agencies and other service providers to ensure the kind of interconnectivity necessary in today's marketplace and in international communications .

Types of standards

- De facto (by convention):that have not been approved by an organized body but have been adopted as standards through widespread use.it is establish originally by manufacturer who seek to define the functionality of a new product or technology.
- De jure (by law) : that have been legislated by an officially recognized body

Standards organizations

- ISO(International organiztions fot standards)
- ITU-T(International Telecoominication Union-Telecommunication)
- ANSI(American National Standard Institute